



BACKGROUND

- At listing, candidates are assigned statuses from 1-6
 - 1 is the most urgent, 6 is the least urgent
- Candidates reach status 2 through either direct entry at listing or transitioning from a different status
- Variation in status 2 entry pathways may contribute to overcrowding within status 2

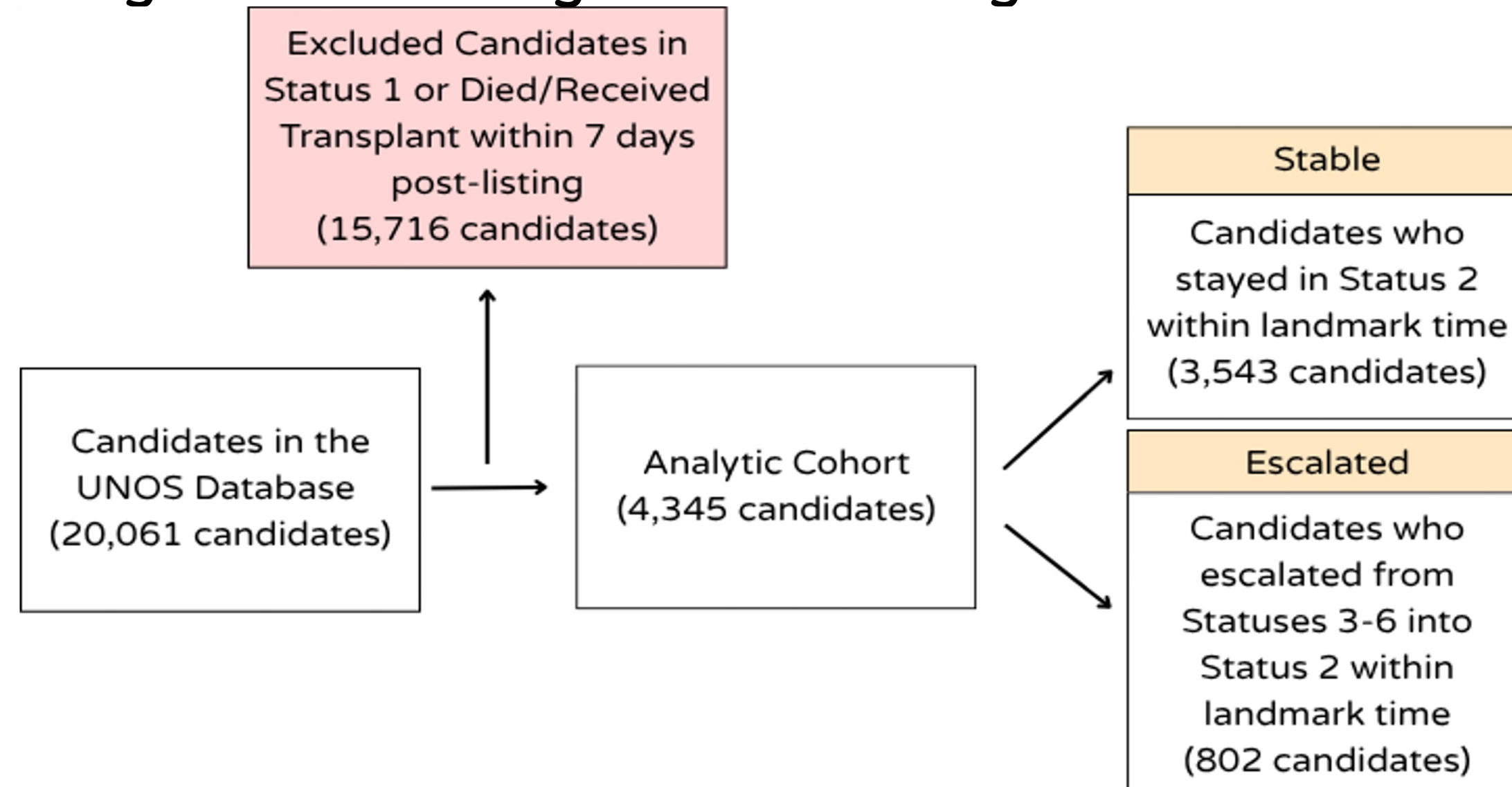
OBJECTIVE

To identify differences in outcomes (death and transplantation) between stable and escalated patients

METHODS

- Used United Network for Organ Sharing (UNOS) data on heart transplant candidates listed from 2018-2024
- Defined a 7-day post-listing landmark time and a 37-day post-listing horizon time for the analytic period
- Fit cause-specific Cox, Fine-Gray, and direct binomial regression models to estimate hazard ratios, subdistribution hazard ratios, and odds ratios

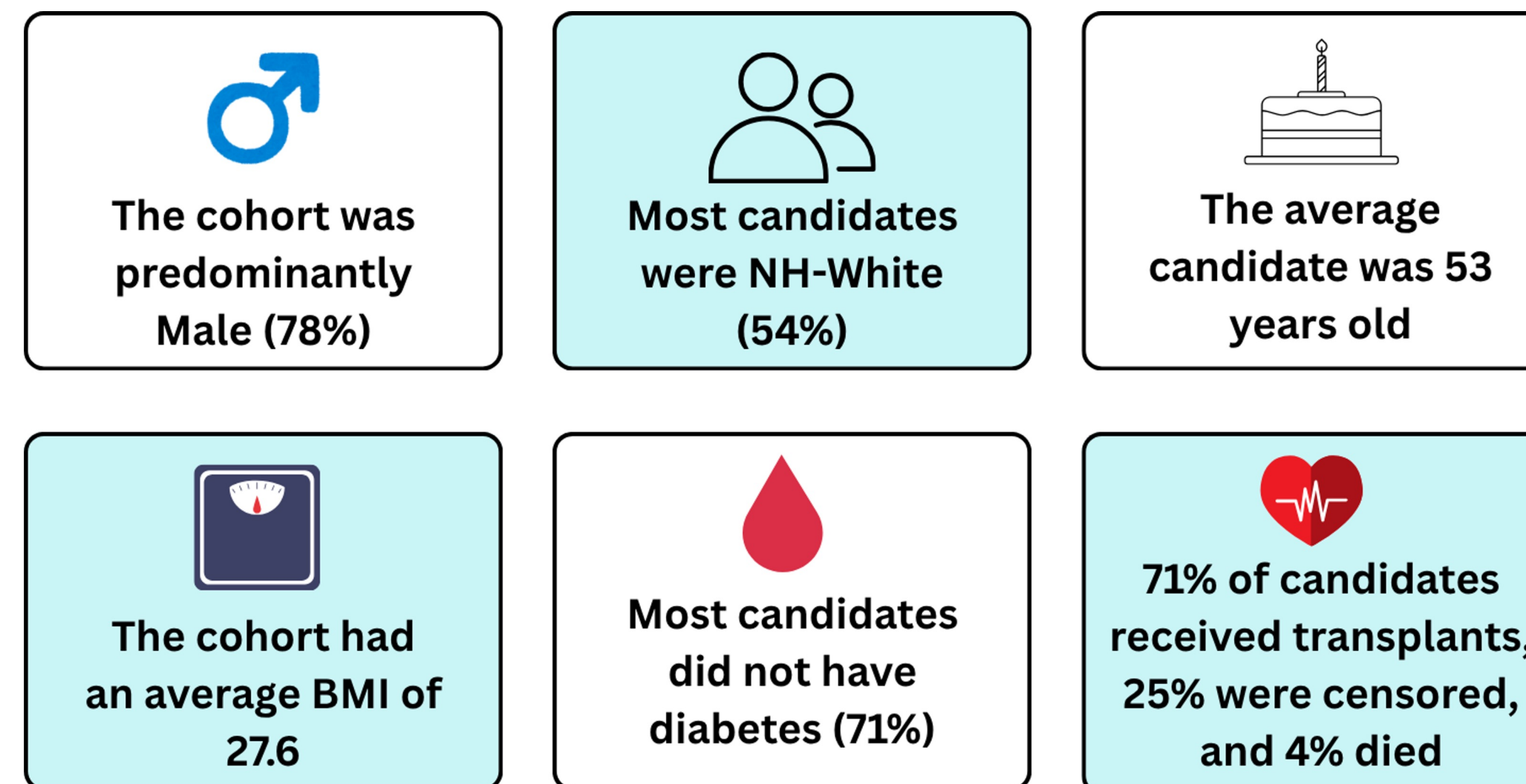
Figure 1. Flow Diagram Showcasing Cohort Creation



RESULTS

Candidate Demographics (n = 4,345)

Figure 2. Summary of Descriptive Statistics



Stable candidates had a higher percentage of exceptions (38%) than escalated candidates (12%).

Analytic Findings

Figure 3. Cumulative Incidence Functions for Transplantation and Death

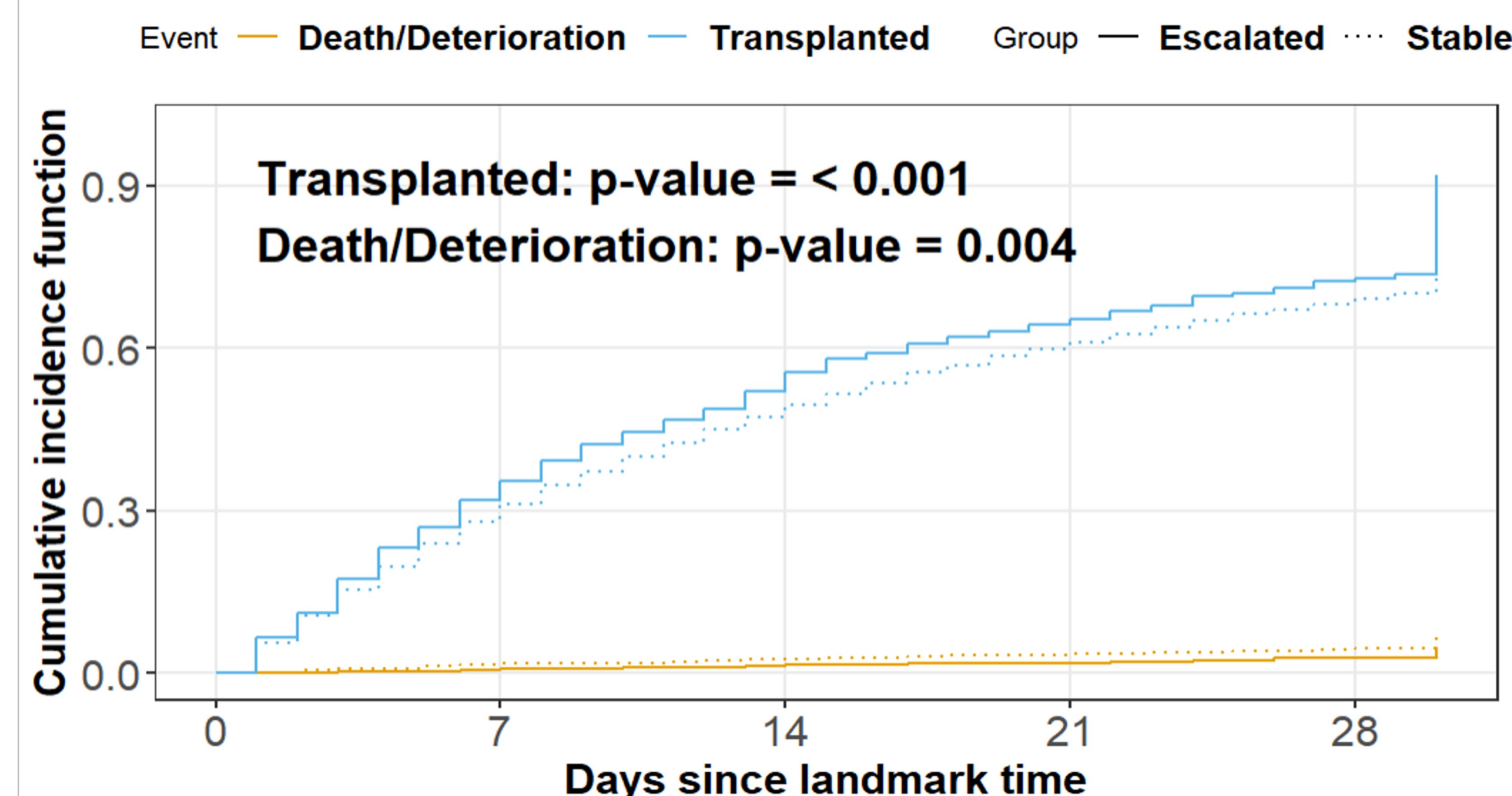


Table 1. Association of Escalation Status with 30 Day Outcomes

Outcome	Estimate**	95% CI	p-value
Cause-Specific Cox Model (HR*)			
Death / Deterioration	0.57	(0.36 , 0.9)	0.015
Transplant	1.12	(1.03 , 1.23)	0.012
Fine-Gray Subdistribution Model (SHR*)			
Death / Deterioration	0.53	(0.34 , 0.83)	0.005
Transplant	1.14	(1.04 , 1.25)	0.004
Direct Binomial Regression Model (OR*)			
Death / Deterioration	0.60	(0.38 , 0.96)	0.034
Transplant	1.19	(0.98 , 1.44)	0.076

*HR = Hazard Ratio; SHR = Sub-Distribution Hazard Ratio; OR = Odds Ratio

**Adjusted for gender, age, BMI, ethnicity, blood type, diabetes, and exception status

Reference group: stable candidates

MAIN FINDINGS

- Escalated candidates have a 43% lower instantaneous rate of death and a 12% higher instantaneous rate of transplantation than stable candidates.
- At day 30, the odds of death are 40% lower for escalated candidates than stable candidates.
- The odds of transplant are 19% higher for escalated candidates compared to stable candidates, but the association was not significant.

DISCUSSION

This study found that escalated candidates had a lower rate of death and higher rate of transplantation than stable candidates.

Limitations

- The 7-day landmark was arbitrary and may bias results by excluding the most urgent Status 2 candidates
- Observational design prevents causal inference

Future Steps

- Future studies should test different landmark times, investigate the effect of exception status on outcomes, and define cohorts differently